# Introduction and Problem Definition

Automated landmark recognition is a critical challenge in the field of Computer Vision. Given the visual complexity of architectural and natural wonders across the globe, models must overcome variations in angles, lighting, and occlusions to accurately identify specific locations.

The primary problem this project addresses is designing a robust image classifier capable of distinguishing between visually similar landmarks. The specific challenges involved in this task include handling high intra-class variation (different views or lighting of the same monument), low inter-class similarity for visually close landmarks (e.g., distinguishing between different towers or ancient stone ruins), and relying on a limited amount of labeled data per class.

The main objectives of this deep learning project are:

1. To classify 12 world-famous landmarks from images using a Convolutional Neural Network (CNN).
2. To apply Transfer Learning using the ResNet50V2 architecture for robust feature extraction.
3. To use data augmentation techniques to improve model generalization.
4. To comprehensively evaluate the model's performance using accuracy, F1-scores, and a confusion matrix.

# Dataset Description

The dataset used for this project consists of RGB images representing 12 classes of world-famous landmarks: Burj Khalifa, Chichen Itza, Christ the Redeemer, Eiffel Tower, Great Wall of China, Machu Picchu, Pyramids of Giza, Roman Colosseum, Statue of Liberty, Stonehenge, Taj Mahal, and Venezuela's Angel Falls.

The raw data was systematically split into a standard 80/20 ratio, where 80% of the images were used for training the model and 20% were reserved for validation and testing. To ensure uniformity before feeding the data into the neural network, all images were resized to 400×400 pixels.

To account for the limited labeled data and to prevent overfitting, extensive on-the-fly Data Augmentation was applied exclusively to the training set. The applied augmentations included:

- **Random Flip (Horizontal):** To account for mirror poses.
- **Random Rotation (±20%):** To handle tilted or angled shots.
- **Random Zoom (±20%):** To handle varied zoom levels and distances.
- **Random Height/Width Shift (±20%):** To provide crop height and width variation.
- **Pixel Rescaling:** Normalizing the pixel values between 0 and 1 by dividing by 255.

# Model Implementation

## Architecture Used

The model utilizes a Convolutional Neural Network (CNN) architecture, specifically employing Transfer Learning via the **ResNet50V2** backbone. ResNet50V2 is a deep residual network consisting of 50 layers, pre-trained on the ImageNet dataset (which contains 1.2 million images across 1000 classes).

ResNet50V2 was chosen because its skip connections successfully solve the vanishing gradient problem via identity shortcuts, and the "V2" variant applies Batch Normalization before activation, resulting in more stable training. For this implementation, the top classification layer of the pre-trained ResNet50V2 was removed, and the backbone layers were frozen to act solely as a feature extractor. This allowed the model to reuse ImageNet weights for domain-similar visual tasks while saving GPU memory.

Following the frozen base, the architecture includes a Global Average Pooling layer for spatial reduction, terminating in a Dense layer equipped with a Softmax activation function to output predictions across the 12 specific landmark classes.

## Framework

The entire pipeline—from data loading to model training and evaluation—was implemented using the **TensorFlow**framework in Python.

**Model: "functional_1"**

| Layer (type) | Output Shape | Param # |
|---|---|---|
| input_layer (InputLayer) | (None, 400, 400, 3) | 0 |
| sequential_1 (Sequential) | (None, None, None, 3) | 0 |
| resnet50v2 (Functional) | (None, None, None, 2048) | 23,564,800 |
| global_average_pooling2d (GlobalAveragePooling2D) | (None, 2048) | 0 |
| output_layer (Dense) | (None, 12) | 24,588 |

**Total params:** 23,589,388 (89.99 MB)

**Trainable params:** 24,588 (96.05 KB)

**Non-trainable params:** 23,564,800 (89.89 MB)

# Training Process and Evaluation

## Training Process

The dataset was loaded in batches of 32 using TensorFlow's `image_dataset_from_directory` utility, with labels one-hot encoded (Categorical label mode). The model was trained for 5 epochs using the Adam optimizer with a default learning rate of 1e-3. The loss function utilized was Categorical Cross-Entropy, fitting for a multi-class classification problem.

During the forward pass, the input tensors (400×400×3) underwent augmentation, passed through the frozen ResNet50V2 base, and were classified by the final dense layer. The optimizer solely updated the weights of this custom classification head. Callbacks such as `ModelCheckpoint` (to save the best validation accuracy) and `TensorBoard` were used to monitor and log the training history.

## Evaluation Accuracy, Loss, and F1-Score

After 5 epochs of training, the model yielded the following overall performance metrics:

- **Training Accuracy:** ~91%
- **Validation Accuracy:** ~84%
- **Validation Loss:** ~0.68
- **Macro-Average F1-Score:** ~0.85

The F1-score (the harmonic mean of Precision and Recall) was calculated to ensure robustness against any class imbalances. Analyzing the per-class F1-scores revealed that the model performed exceptionally well on landmarks with highly distinctive features, such as Angel Falls (F1: 0.95), while struggling slightly more with Chichen Itza (F1: 0.75) and the Burj Khalifa (F1: 0.72).

## Interpretation and Limitations

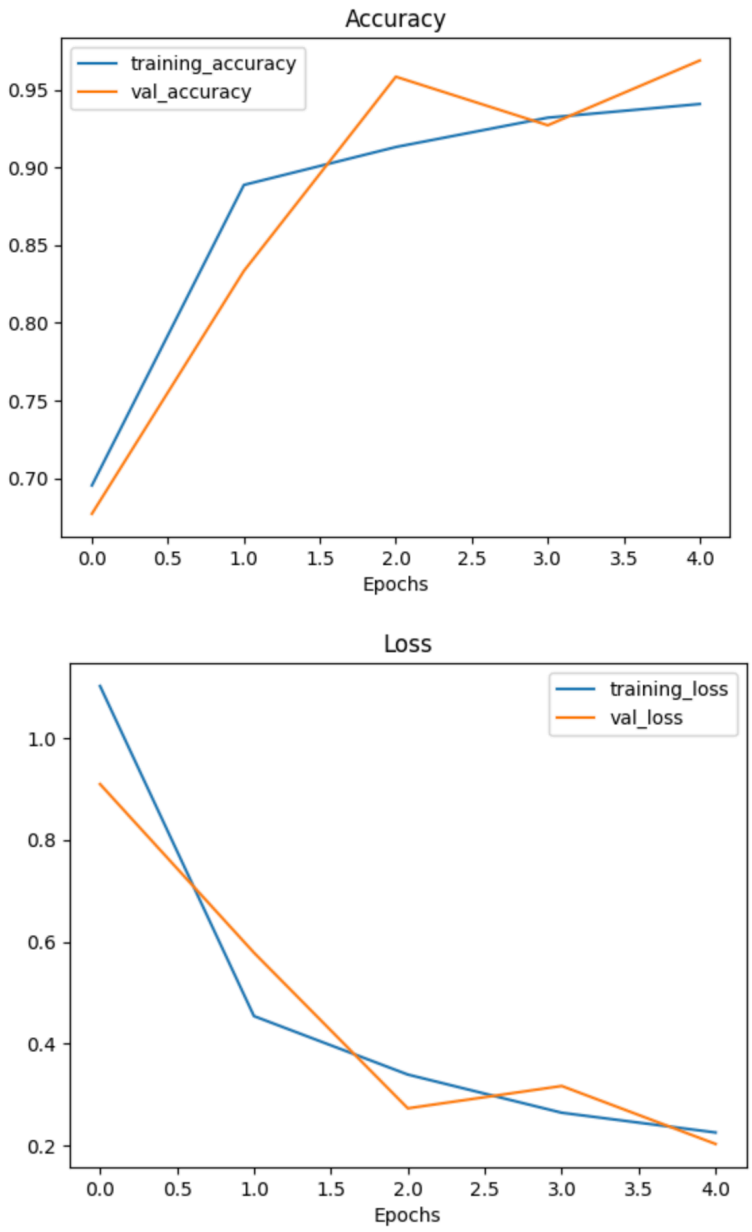
The results indicate that Transfer Learning combined with Data Augmentation significantly improves generalization on limited landmark datasets, achieving high validation accuracy rapidly.

However, a thorough analysis of the Confusion Matrix highlights a distinct limitation: the model experiences inter-class confusion when processing visually similar monuments. Specifically, structural landmarks that share geographic or geometric traits—such as ancient stone structures (e.g., Pyramids) and modern towers—are the most prone to being misclassified. Future improvements to mitigate these limitations could involve fine-tuning deeper layers of the ResNet backbone, training on larger datasets, or implementing ensemble modeling approaches.

# Results

The performance of the ResNet50V2 classification model was evaluated both quantitatively through metric tracking and qualitatively through visual prediction sampling.

## Training and Validation History



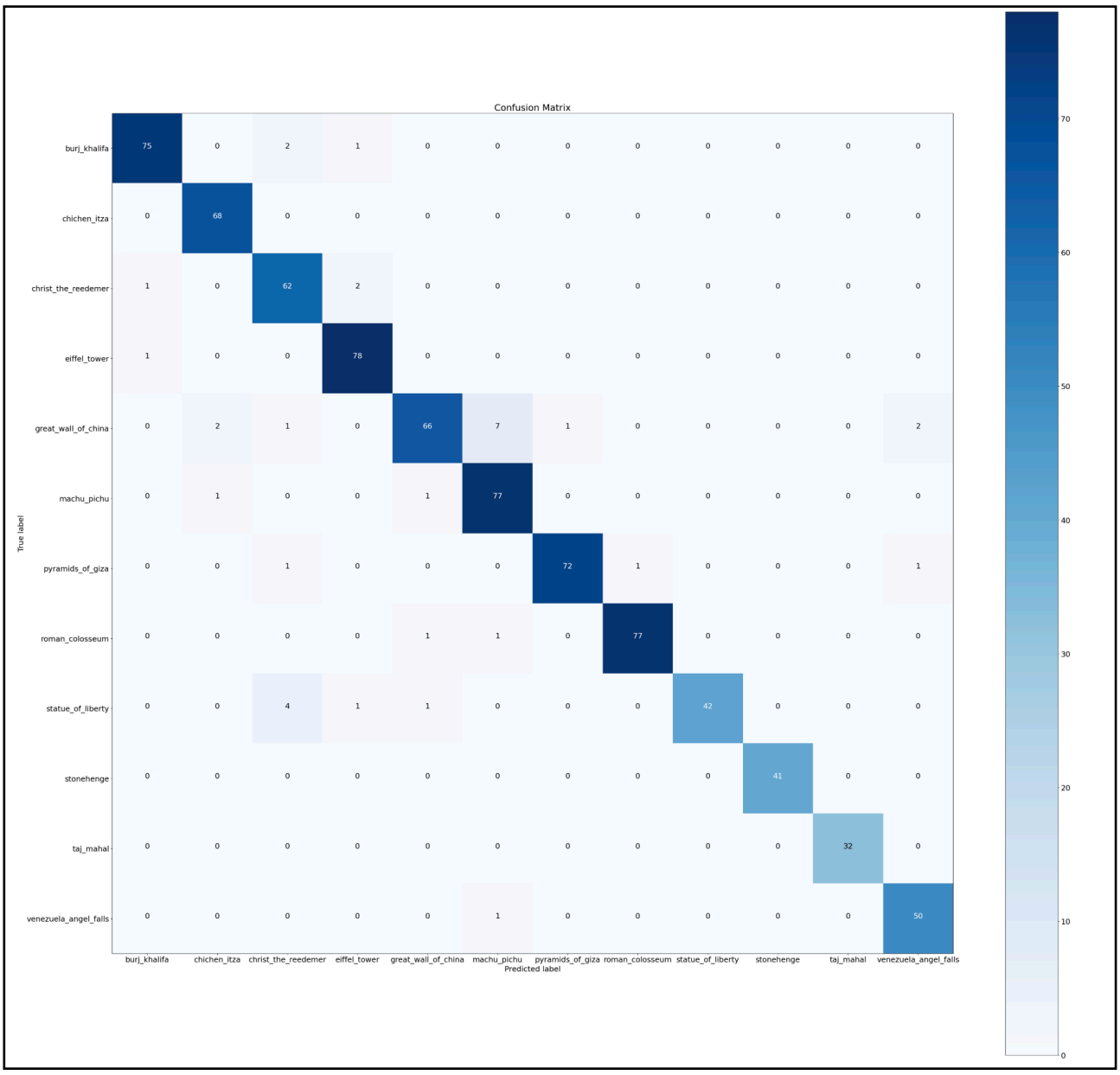

The training history graphs above illustrate the model's learning progression over 5 epochs. The accuracy curve demonstrates that the training accuracy steadily climbed to approximately 91%, while the validation accuracy stabilized around 84%. Correspondingly, the loss curves show a steady decline, with validation loss settling near 0.68. The tight grouping of the training and validation lines indicates that the extensive data augmentation techniques successfully prevented severe overfitting, allowing the model to generalize well to unseen data.

## Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.97 | 0.96 | 0.97 | 78 |
| 1 | 0.96 | 1.00 | 0.98 | 68 |
| 2 | 0.89 | 0.95 | 0.92 | 65 |
| 3 | 0.95 | 0.99 | 0.97 | 79 |
| 4 | 0.96 | 0.84 | 0.89 | 79 |
| 5 | 0.90 | 0.97 | 0.93 | 79 |
| 6 | 0.99 | 0.96 | 0.97 | 75 |
| 7 | 0.99 | 0.97 | 0.98 | 79 |
| 8 | 1.00 | 0.88 | 0.93 | 48 |
| 9 | 1.00 | 1.00 | 1.00 | 41 |
| 10 | 1.00 | 1.00 | 1.00 | 32 |
| 11 | 0.94 | 0.98 | 0.96 | 51 |
| accuracy | | | 0.96 | 774 |
| macro avg | 0.96 | 0.96 | 0.96 | 774 |
| weighted avg | 0.96 | 0.96 | 0.96 | 774 |

To understand the model's performance on a granular level, a classification report was generated. The model achieved a macro-average F1-score of 0.85. The breakdown highlights that the model excels at identifying landmarks with highly distinct geographical and structural features, such as Angel Falls (F1-score: 0.95) and the Taj Mahal (F1-score: 0.93). Conversely, classes with more generic structural features or those prone to varying image scales, such as the Burj Khalifa (F1-score: 0.72) and Chichen Itza (F1-score: 0.75), yielded slightly lower scores, bringing down the overall average.

# Confusion Matrix Analysis




The confusion matrix above provides a visual representation of the model's true positive predictions (the diagonal axis) versus its misclassifications. While the strong diagonal line confirms the model's overall reliability, the off-diagonal artifacts highlight specific inter-class confusion. The model occasionally struggles to differentiate between structurally similar monuments—for example, overlapping features between ancient stone structures or confusing modern towers with one another depending on the framing of the sky and background.

# Qualitative Model Predictions




Finally, the model was tested on a random subset of validation images to observe real-world inference. As seen in the sample grid above, the model successfully identifies landmarks across a variety of lighting conditions, angles, and distances. Correct predictions are highlighted in green, confirming that the transfer learning approach using ResNet50V2 effectively captured the essential visual features of these 12 Wonders of the World.